F = ma in practice

LEARNING TARGET · R7 I can solve with F = ma — always finding the NET force first.

SECTION A KEY VOCABULARY

Net force	Combine opposite forces: subtract and keep the direction of the larger one.
F = ma	Uses the *net* force, never just one pull.
Direction	Acceleration points the same way as the net force.
Balanced	Net force 0 N → a = 0 m/s² (still or constant velocity).

SECTION B KEY EQUATION + UNITS

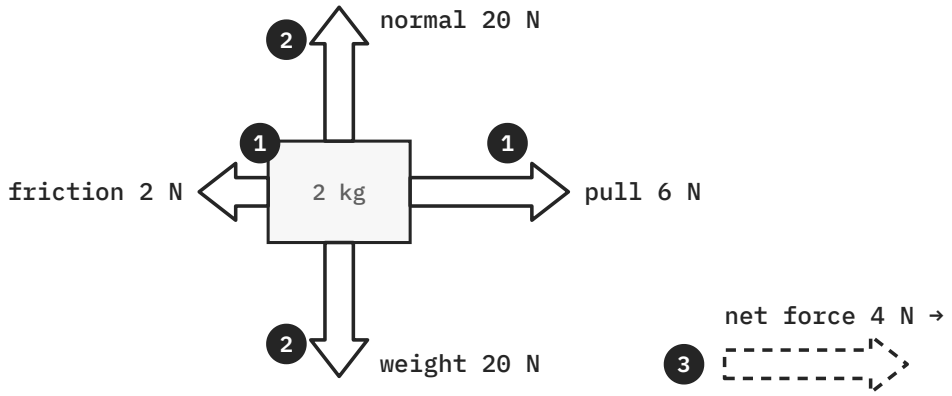
$$\Sigma F_x = m\ a_x$$

$$\Sigma F_x = F_{right} - F_{left}$$

F	newtons (N)
m · a	kg · m/s² (1 N = 1 kg·m/s²)

SECTION C COLOR IT · THE ARROWS COMPETE

Vertical forces balance.



- ☐ **1 · ORANGE** — Horizontal forces
- ☐ **2 · YELLOW** — Vertical forces (balanced)
- ☐ **3 · ORANGE** — Net force (thick arrow)

SECTION D WORKED EXAMPLE

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration.

GIVEN	EQUATION	WORK	ANSWER
m = 4 kg 12 N right 4 N left	$a = F_{net} \div m$	$F_{net} = 12 - 4 = 8$ N right $a = 8 \div 4$	**2 m/s² right** with the net force

SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: F = MA

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Find the net force before dividing by mass

Using Newton's second law with opposing forces

A strong pull does not act alone when friction is also present. To predict acceleration, we need the combined effect of all the forces on the object. Newton's second law uses net force. I want that force sum to appear before the division by mass, so the calculation shows which interactions were included.

Choose an axis and a positive direction. With right positive, a rightward force enters the sum as positive and a leftward force as negative. Along the horizontal axis, $\Sigma F_x = ma_x$. The symbol Σ means sum. For two opposing forces, this becomes $\Sigma F_x = F_{right} - F_{left}$.

Compare the horizontal arrows in **Section C**. They act on a **2 kg cart** pulled right by **6 N**, with **2 N of friction left**. Its horizontal net force is **6 N – 2 N = 4 N** right. The 20 N upward normal force balances the 20 N downward weight. Those vertical forces do not contribute to the horizontal sum, and their own vertical sum is zero.

The thick net-force arrow summarizes the two horizontal forces. It is not an additional interaction to add again. If we counted the pull, friction, and net-force arrow as three separate forces, we would count the same imbalance twice. The diagram and force equation are two ways to describe the same situation.

The example in **Section D** uses a **4 kg cart**, a **12 N pull right**, and **4 N of friction left**. First combine the forces: $F_{net} = 12\ N - 4\ N = +8\ N$. Then use $a = F_{net} \div m$, giving $a = 8\ N \div 4\ kg = 2\ m/s^2$ right. Dividing 12 N by 4 kg would ignore friction.

Acceleration points in the direction of the net force. Its size tells us the velocity change each second. The unit check agrees: $1\ N = 1\ kg \cdot m/s^2$, so newtons divided by kilograms give m/s². We can also check backward: $(4\ kg)(2\ m/s^2) = 8\ N$, matching the net force we found.

A rightward acceleration does not prove the cart is moving right. If it already moves right, it speeds up. If it moves left, it initially slows down. We need its starting velocity to describe that part of the motion. If the forces balance, acceleration is zero; the cart can remain still or keep a constant velocity.

For our fictional asteroid, 2026-XJ, mass and net force determine acceleration. Then the existing velocity and elapsed time tell us how that acceleration changes the motion. A careful force sum is the first link in that chain. It keeps a missing or double-counted interaction from changing the entire forecast.